HUMBER POLYTECHNIC | Faculty of Applied Sciences & Technology

ENGI 4000

CAPSTONE PROJECT 1

CHAPTER 2

ARTICULATE ENGINEERING DESIGN PROBLEM WITH PARAMETERS AND NEEDS OF PERSPECTIVE STAKEHOLDERS





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Executive Summary

This chapter serves to prepare a mission statement for the engineering design problem for the High-Speed Camera Robot. It shows who the product would be for, what similar products already exists, where the gaps are, what they provide, what we potentially would be providing, and what our team essentially can build. This report can be broken down to three main sections being the market research and competitive solutions, the development of opportunities, and the mission statement which is what this report is primarily arriving to.

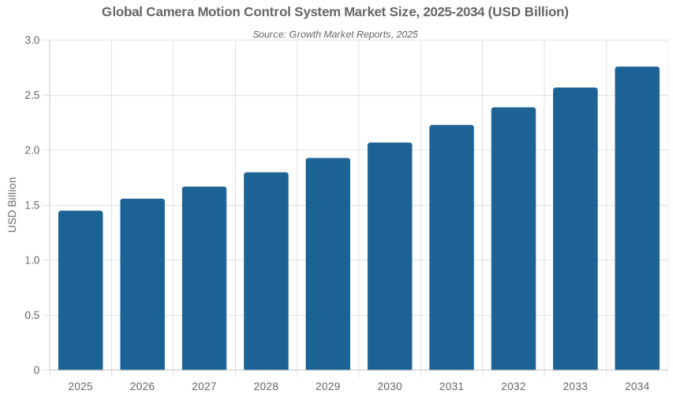
Since the client wants a programmable camera-motion system for high-speed cinematography, visual-effects shots, stop-motion shots, time-lapse and other moves that are hard to do with ordinary equipment, a section of this report serves to identify who would be in the market looking for this type of product to solve those issues and current existing competitions with this product. Each would be compared and analyzed to find what advantage the prototype can have over these products.

Research estimated the value of the camera motion control market at about US$1.3–1.45 billion in 2025, growing roughly 7–9% a year hence the varying competitive businesses that grew with many different features as listed in the report. This report provides a project charter and 12 opportunities, each with details and diagram/sketch of design. The mission statement will serve to define the product, its benefit, business goals, target markets, assumptions/risks/constraints and stakeholders.

A. Market Research and Competitive Solutions

A.1 Market Overview and Segments

Looking at the market for this product, it can be seen that the estimate of the global camera motion control system market is at around US$1.3–1.45 billion in 2025, with forecast growth of about 6.7% to 9.5% per year depending on the source according to market reports.



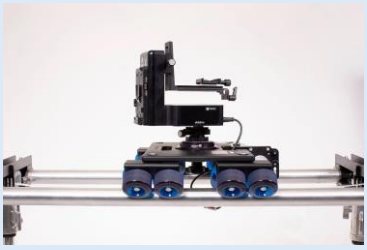
The market segments that would mainly be included with this product would be:

- Film, TV and commercial production
- Product and tabletop photography studios
- Stop-motion animation and time-lapse creators
- Education, research and broadcast or live events

Segment	Needs	Expectations
Film, TV and commercial production	Product can be used for high-speed shots, VFX plates, repeatable passes for scene shots, ad shots, etc.	Repeatability, speed, low vibration, camera mount compatibility
Product and tabletop photography studios	Product can be used for macro, product orbiting and reveal shots in this segment	Precision, easy programming, portability
Stop-motion animation and time-lapse creators	Can be used for frame-by-frame movement and moving time-lapse	Repeatability, trigger software integration, low cost, sturdy build
Education, research and broadcast / live events	Can also be for film-school labs, demonstrations, robotic studio cameras	Safe, easy to use, able to document and affordability

A.2 Competitive Product Comparison

Product	Type	Payload	Performance
MRMC Bolt 	6-axis high-speed arm; optional track and PTR head	Up to 20 kg	Can go up to 12 m/s with track,180° rotation in 1 s
MRMC Bolt Mini Model Mover 	6-axis arm, 42 kg, single-phase power	Up to 8 kg	Has a ±0.02 mm tolerance on repeatability, 270°/s axis speed, 10.5 m/s max combined speed
MRMC Bolt Jr+ 	Compact 6-axis arm	Up to 10 kg	Longer reach than Bolt Jr; set up in under an hour
Motorized Precision EVO / KIRA 	Cinema robot arms	Up to 5 Kg	KIRA has a ±0.05 mm tolerance repeatability, EVO has hand-guided path recording

Product	Type	Payload	Performance
eMotimo Spectrum ST4 / ST4.3 	Pan/tilt head, 4 axes (up to 7 with expansion) plus slider	Up to 7 kg	Can go up to 7 keyframes and 11 preset moves (ST4)
Kessler CineShooter+ with CineSlider 	Pan/tilt head + motorized slider, multi-axis	Up to 8 Kg	Gearbox lash under 0.01°,Unreal Engine integration
edelkrone SliderPLUS v5/v6 + HeadPLUS 	Motorized slider + modular heads, app control	Up to 20 Kg	Multi-axis sync in one app, Point Tracking, motion time-lapse, stop motion, VFX

Family	Key features	Benefit to the user
Robot-arm cinebots (MRMC Bolt family, Motorized Precision)	Six axes; very high speed, has 0.1 mm repeatability, cinema software, external trigger sync.	Extreme, complex moves, VFX-grade repeat passes, model-mover options
Professional heads and sliders (Kessler, eMotimo)	Keyframe and preset moves, focus/iris/zoom motors, Dragonframe integration, about 15 lb payload.	Affordable, portable, proven for time-lapse, interviews, product and stop-motion work
Prosumer modular (edelkrone)	App-controlled sliders and heads that sync; Point Tracking; motorized add-ons.	Very portable and low cost, fast to set up

A.3 Customer Needs and Expectations

The needs and expectations are as shown in the table below:

Customer need	Design expectations
Exactly repeatable moves	Closed-loop control, homing, low-backlash drive,
Fast moves for 240+ fps slow motion	High acceleration and speed with adequate motor torque and a stiff structure
Vibration-free motion	Stiffness, damping, jerk-limits, IMU vibration measurement
Moves impossible with standard gear	Programmable multi-axis trajectories
Easy programming and replay	Keyframe interface with preview, save and replay
Camera and light synchronization	Trigger output with timecode tied to the camera position
Safe operation near people	E-stop, limit switches, guarding
Portability and affordability	Modular design, standard mounts, cost much less than robot arms

A.4 Competitive Advantages and Differentiation

Our competitive advantage over other companies is as follows:

Advantage	Differentiation
Much lower cost than an arm	Needs to only use three to four powered axes instead of six which reduces the amount of materials needed in comparison to other companies.
Rigid, simple structure	A linear rail is much easier and simpler to control than a long multi-joint arm which is a more complex build. Has better repeatability and vibration behaviour control as well.
Safety by design	Has bounded linear envelope, hard limits, hardware E-stop chain which makes it perfect for educational use or being new user friendly.
Modular and interchangeable	Quick-change camera mount, optional modular rail sections



Image showing concept design of intended product.

B. Development of Opportunities (DES1)

B.1 Project Charter

Project title	High Speed Camera Robot
Charter statement	This project is a programmable camera-motion platform that moves a camera quickly, smoothly and repeatably along a motorized linear rail with rotary camera axes.
Purpose / business case	The purpose of this prototype is to give film, VFX, stop-motion and photographers a way to make high-speed, repeatable and difficult camera maneuvers without the need for a costly and complex robotic-arm system.
Objectives	• Mechanically rigid camera-motion platform suited to camera mounting • Programmable, repeatable movements with high accuracy • Physical limits and an emergency stop for safe operation • Has good speed, acceleration, vibrational control and repeatability • Has an interface that supports camera movement
Scope	• In scope: motorized linear rail, rotary axis, camera mount, motion-tracking vision, high-speed movement • Out of scope: robotic arm, waterproofing, heavy payloads, unrestricted outdoor operation
Key milestones	Project acknowledgement, project management plan, mission statement, proposal, concept, risk analysis, preliminary design, model and simulation, design review, finalized design, presentation and report, prototype assembly, motion-control and safety testing, trajectory validation, final documentation and presentation.
Team and roles	Joshua Mongal – project coordination, part design, modelling and simulation Mikaeel Khanzada – controls, software integration, interface Michael McCorkell – fabrication, wiring, assembly and testing
Sponsor / key stakeholders	Faculty of Media, Creative Arts and Design students and director Matthew Kennedy.
Success criteria	Should perform repeatable trajectory executed with minimal deviation, have stable position/velocity following, have E-stop and travel limits stop to prevent accidents, camera stays secured and minimal shake affecting the scene shot
Constraints and assumptions	Time, budget, component availability, machining/3D-printing access, camera type, speed and workspace rules
Approval	Supervisor: _____ Date: _____

B.2 Opportunity Generation and Overview

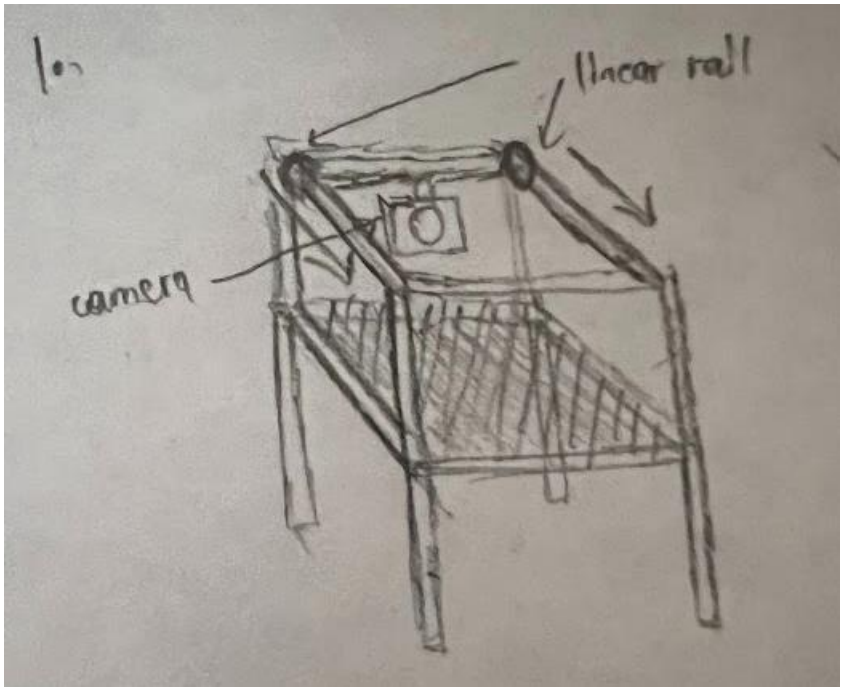
The List of opportunities are as follows in the table below.

1	High-speed linear rail carriage	Fast, straight, large travel
2	Pan-tilt-roll rotary head	Camera angle changes while moving
3	Vibration damping and stiffness	Shake magnified in slow motion
4	Closed-loop feedback and repeatability test	Exactly repeatable passes
5	Jerk-limited trajectory planner	Fast moves without shake
6	Shot programming interface	Programming without coding
7	Camera trigger and timecode sync	Camera events tied to motion
8	Motor-current monitoring	Detect jams and collisions
9	Layered safety system	Fast mass near people
10	Quick-change camera mount	Swap cameras, balance loads
11	Vision-based subject tracking	Follow moving subjects
12	Modular portable rail	Different studio sizes; transport

B.3 Opportunity Schematics

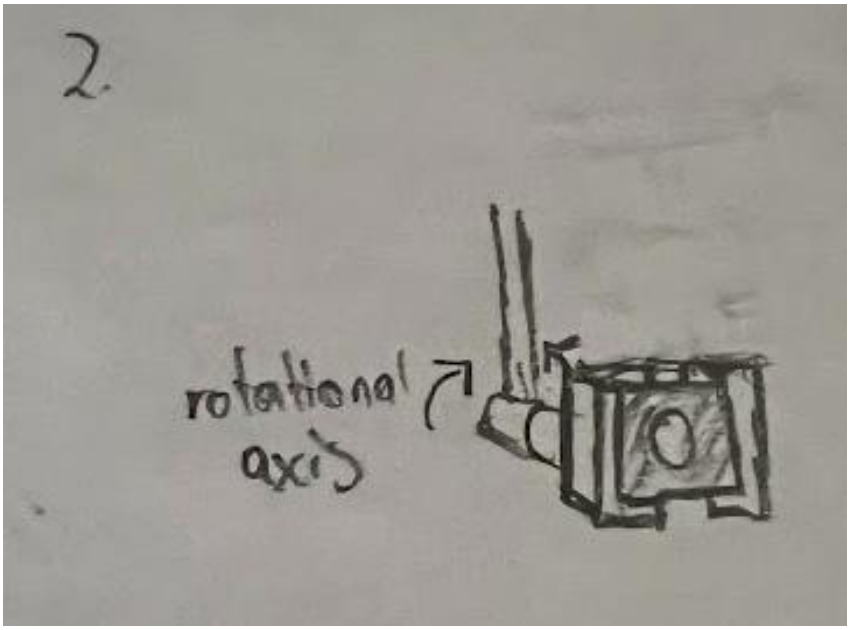
B.3.1 Opportunity 1: High-Speed Linear Rail Carriage

Need	The client wants fast, smooth camera moves that are hard to repeat consistently by hand, especially at 240 fps and above. Any variation in speed or acceleration shows up in the footage.
Idea	A rigid rail with dual guides and a servo-driven carriage that carries the camera and rotary head. Hard end stops protect both ends.
Expected benefit	This is the large-movement axis that is in the project plan. One long linear axis has simple kinematics and no singularities when compared to a six-axis arm.
Main challenge	Belt stretch or ball-screw whip at long travel; carriage mass limits acceleration; the carriage must stop safely in the available distance.



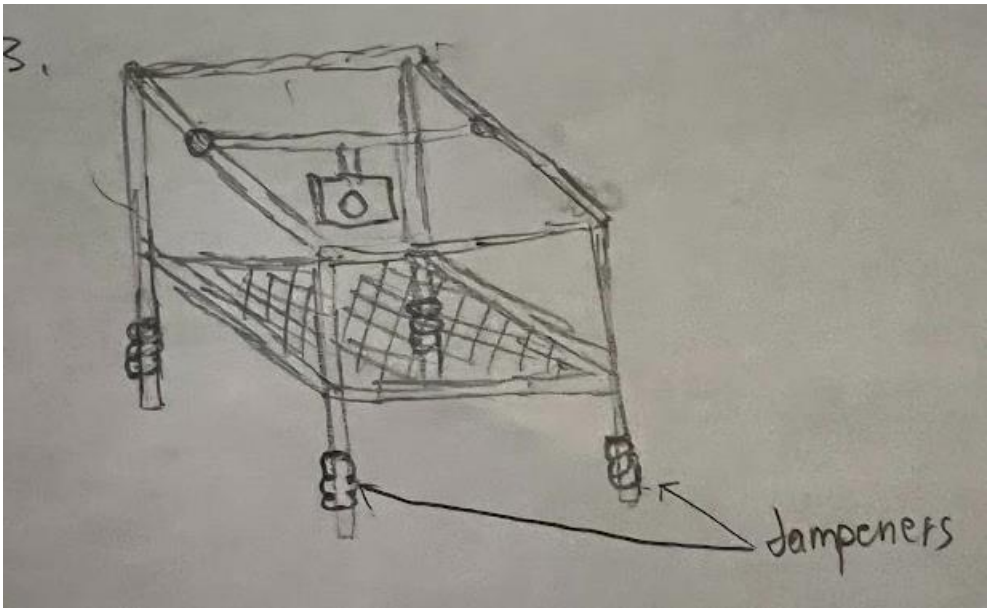
B.3.2 Opportunity 2: Pan-Tilt-Roll Rotary Head

Need	A camera mount that can keep a subject framed or change angle during a shot. Reveals, tracking and creative angles need rotation where camera is mounted.
Idea	A compact two axis mount for the camera, driven by gearboxes, with axes that can intersect near the camera's mounting point, so rotation can occur for different angles
Expected benefit	Keeps the subject in frame during rail motion and adds angles a normal mount can't cannot produce.
Main challenge	Extra moving mass and cable routing on the mount would be complex, possible backlash and inertia matching, also motor torque needed to hold an unbalanced camera may not be enough.



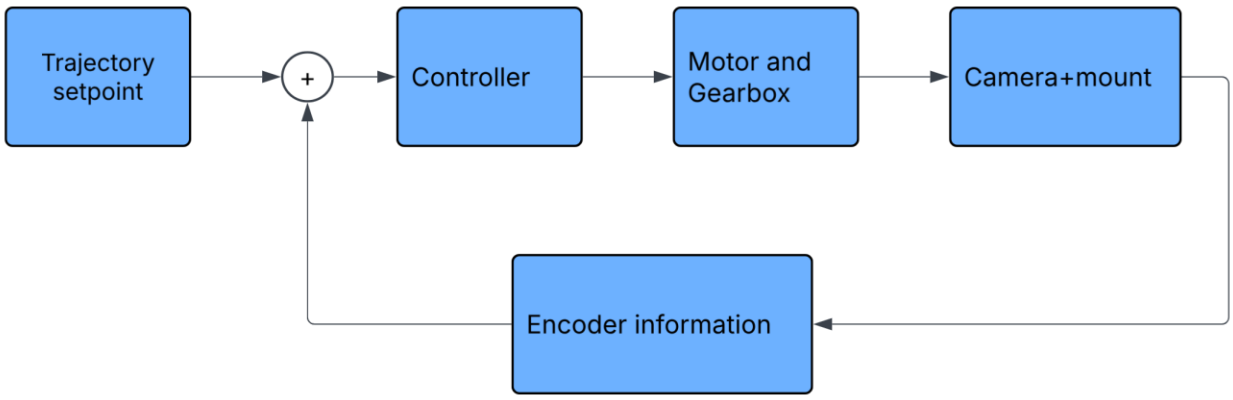
B.3.3 Opportunity 3: Vibration Damping and Stiffness Optimization

Need	Most scenes taken requires system to not be shaking as it ruins the shots.
Idea	A closed-section, high-stiffness beam on spring-damper isolation feet for system, with an IMU on the carriage to measure resonance.
Expected benefit	Shorter settling time after stops and visibly cleaner footage, backed by measured data.
Main challenge	Added mass from damping hardware can affect speed.



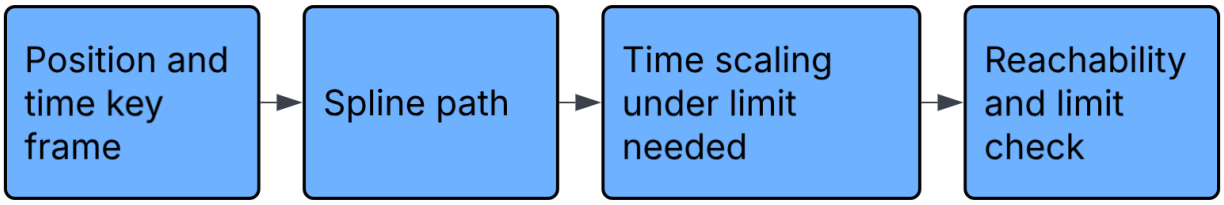
B.3.4 Opportunity 4: Closed-Loop Feedback and Repeatability Verification

Need	The client asks for accurate repeatable motion in replicating shots.
Idea	Motor encoder plus an independent linear scale, homing at start-up, and PID for correct system adjustments to reproduce movements well.
Expected benefit	Turns repeatability from a claim into a number within a small margin of error with auto correction.
Main challenge	Small tolerance measurement on accuracy would need a good reference or record of sequence it took.



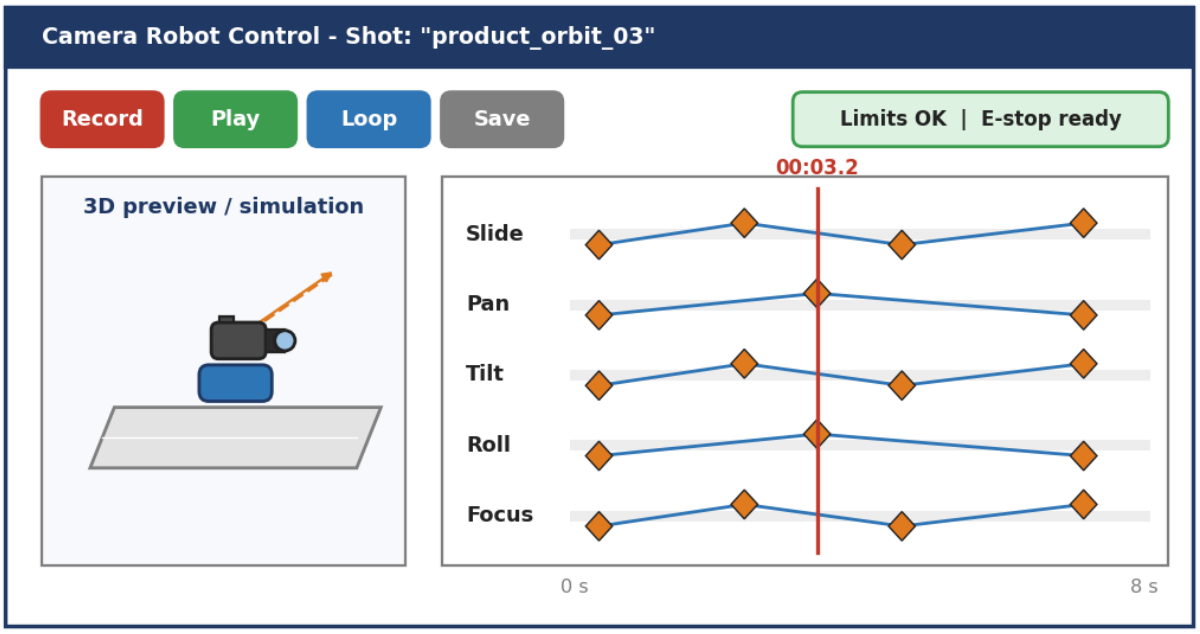
B.3.5 Opportunity 5: Jerk-Limited Trajectory Planner

Need	Since fast movement can cause vibration when acceleration changes abruptly. Repeatable timing needs deterministic motion.
Idea	Can convert movement to fix into the motion of a spline path with jerk limits, can also check reachability and limits, then stream the synced setpoints to all axes to ensure no resonance occurs.
Expected benefit	Shorter efficient moves without shake, and identical motion on every run.
Main challenge	Real-time execution requires fast response/feedback.



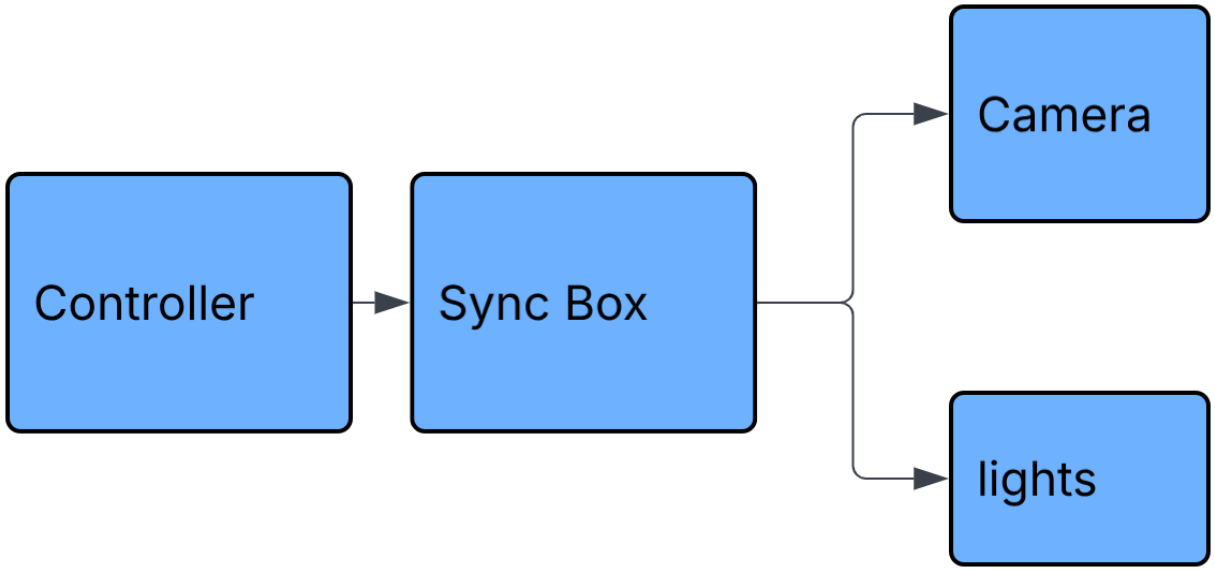
B.3.6 Opportunity 6: Shot Programming Interface

Need	Cinematographers and animators need to be able to program the sequences, preview and repeat a move quickly without writing a complex code.
Idea	Can add a timeline interface with keyframes per axis, a simulated preview before execution, save/load/replay of shot files, speed override and live stop status.
Expected benefit	Fewer trial passes needed, the preview catches limit violations before the hardware moves to prevent accidents, shots are more reproducible.
Main challenge	Interface development time, reliable communication with the controller.



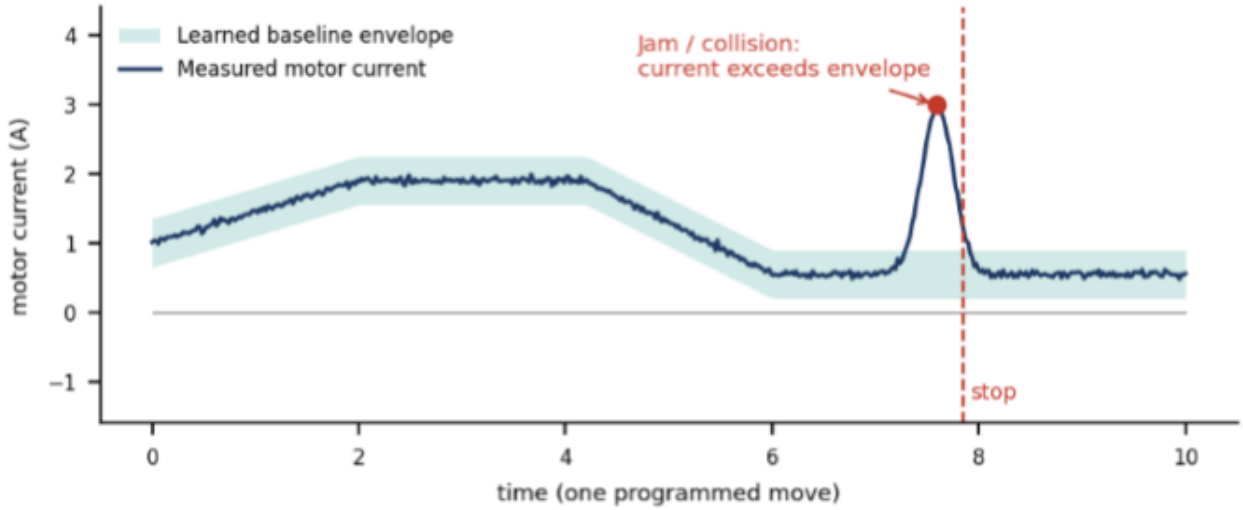
B.3.7 Opportunity 7: Camera Trigger and Timecode Synchronization

Need	Features of the camera including trigger can be done from system without reliability on the camera itself.
Idea	A sync box added to system for triggering record or photography features.
Expected benefit	Removes the need to interact with the actual camera device and access main functions through the system itself.
Main challenge	Camera interfaces vary by model, and the camera latency may interfere with the response times needed for scene.



B.3.8 Opportunity 8: Motor-Current Monitoring and Fault Detection

Need	Jams, collisions and wear should be detected without adding many sensors.
Idea	Sample drive current recorded during each move and compare it with a baseline run, and command a controlled stop when the current spike more than expected when detected
Expected benefit	Low-cost protection using data the drive already has, plus diagnostic data for validation.
Main challenge	False trips at high acceleration can activate stop and interfere with shots.



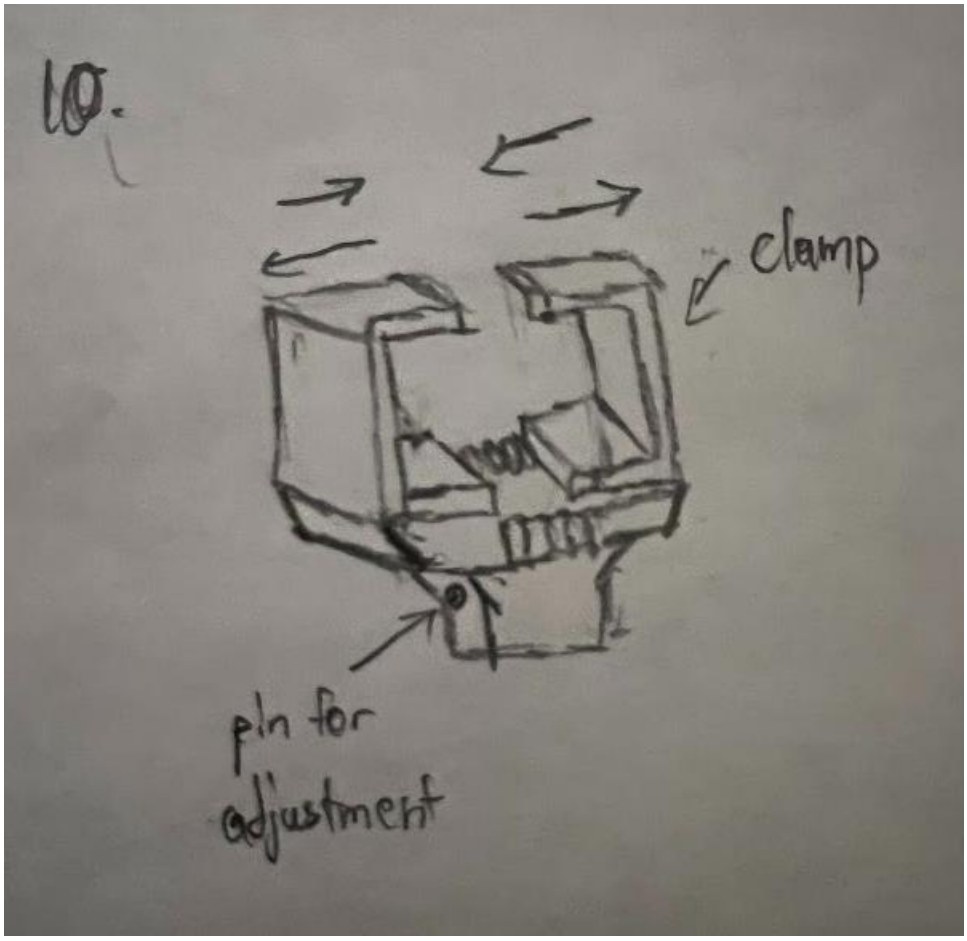
B.3.9 Opportunity 9: Layered Safety System

Need	A fast-moving mass near people is the main hazard hence a system to automatically stop when a presence is in the way is needed to prevent collision.
Idea	E stop with proximity sensor can be added to prevent collision.
Expected benefit	Safe testing and demonstration, protection that works even if software fails.
Main challenge	Stopping distance at maximum speed without damage.



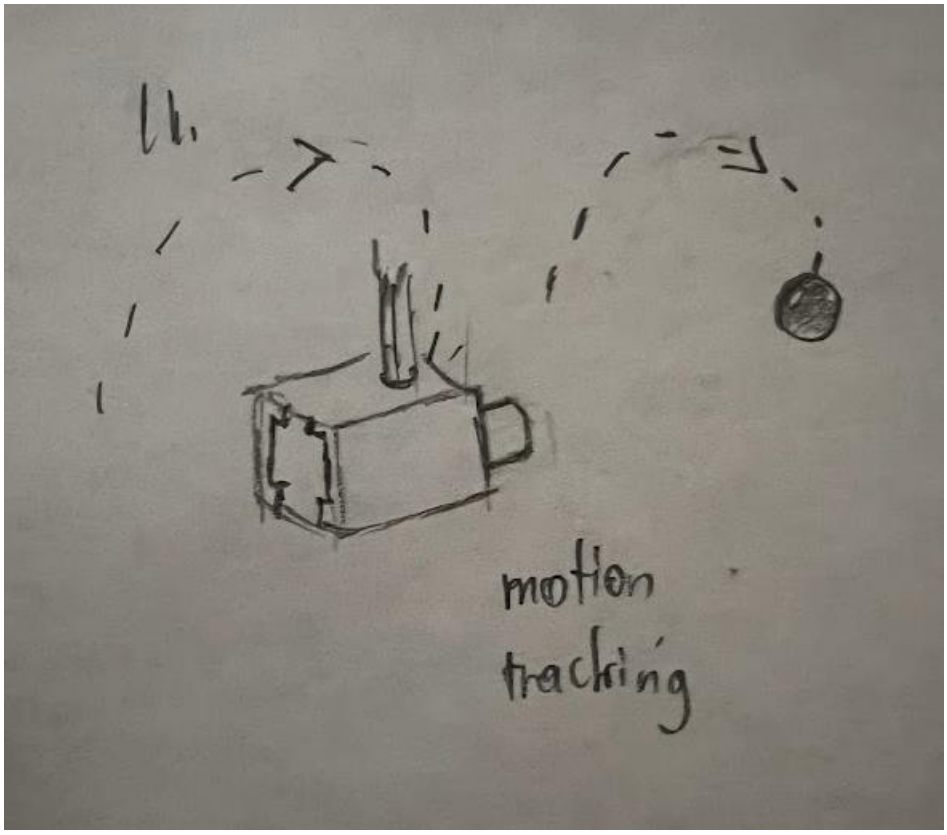
B.3.10 Opportunity 10: Quick-Change Camera Mount with Balance Adjustment

Need	Different cameras and lenses will be used, and imbalance raises motor loads and vibration. The camera must stay secured throughout the move while maintaining balance.
Idea	A standard dovetail plate with slotted balance adjustment, a tool-free locking clamp, alignment pins for repeatable seating.
Expected benefit	Fast camera swaps, repeatable seating and a check that the camera is locked before motion.
Main challenge	Can add to vibration, extra mass and limit type of cameras to be mounted.



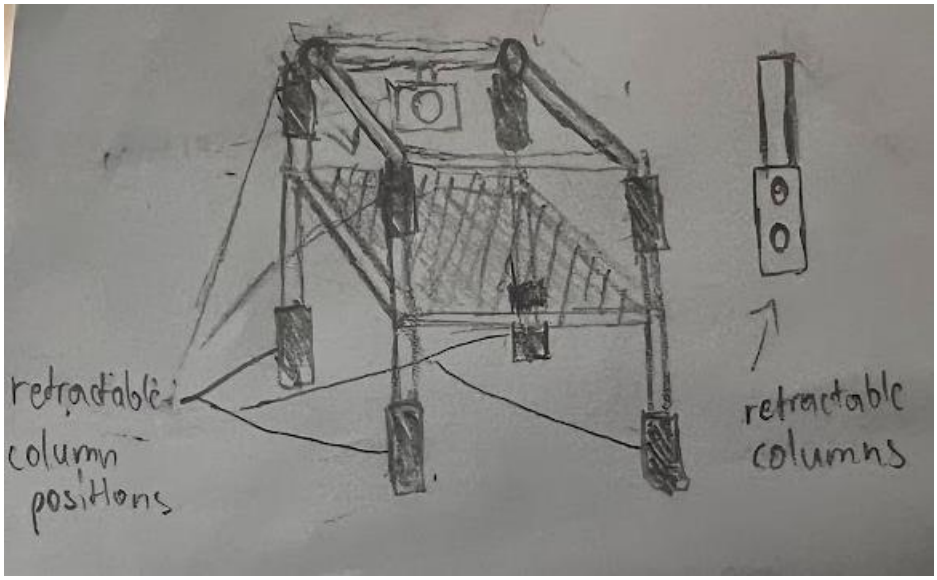
B.3.11 Opportunity 11: Vision-Based Subject Tracking

Need	Some shots would need to follow a moving subject or product without a pre-planned path.
Idea	Detect a target in the camera image, convert its pixel offset to pan/tilt rate commands with a PID controller, and optionally drive the motor to move camera according to the path needed.
Expected benefit	Automatic framing of moving subjects, the vision link can also help verify repeatability.
Main challenge	Image-processing latency, lighting, computing load can get in the way of motion tracking



B.3.12 Opportunity 12: Modular Portable Rail System

Need	Studios differ in size, system needs to be portable to be set up in any location
Idea	Build sections can have push pin adjusters to retract columns for ease to carry or store
Expected benefit	Reconfigurable column length and easier transport and setup.
Main challenge	May interfere with wiring or cause faulty sensors



C. Mission Statement

Product description	A programmable camera-motion platform that carries a camera through fast, smooth and exactly repeatable movements.
Benefit proposition	• Repeatable passes without deviation • Fast, low-vibration motion suited to 240+ fps slow-motion capture • Programmable moves that are difficult or impossible with tripods, dollies and gimbals • Built-in safety for accident-free performance • A fraction of the cost and less complexity to that of a robotic-arm system
Key business goals	• Time: finalize the design and build prototype within 28 weeks • Cost: prototype bill of materials within the budget approved by the supervisor • Quality: meet the acceptance targets for repeatability, motion control, safety stop, secure camera mounting and minimal vibration • Impact: Able to be demonstrated and used by anyone.
Target market(s)	Target markets include independent and small-studio films, commercial and VFX productions, product and tabletop photography studios, stop-motion and time-lapse creators
Assumptions, risks and constraints	Assumptions: suitable workspace and equipment, final camera and payload chosen, e-stops/ auto stop features, and multiple axes should be used Risks: vibration at high speed, component acquisition times, budget overrun and camera not finalized/compatible. Constraints: time schedule, budget, component availability, access to machining and 3D printing, camera type, achievable speed, workspace rules and policies
Stakeholders	• End users: cinematographers, VFX artists, stop-motion animators, product photographers • Faculty of Media, Creative Arts and Design (Matthew Kennedy) • Component suppliers

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